Independent Reproduction of Adaptive Directed Graph Retrieval for Vietnamese Medical Question Answering

Technical reproduction report (paper submitted to *Applied Intelligence*)

October 12, 2026

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1 Executive summary

The paper proposes ADGR: query-aware retrieval over a Vietnamese medical knowledge graph (hybrid anchors, semantic edge filtering, four-source aggregation, cross-encoder reranking). We rebuilt everything from scratch on one NVIDIA A30: 1,934 crawled articles, 15,837 chunks, 52,578 entities / 106,279 relations, 300 questions × 5 pipeline variants, two LightRAG baselines, official-algorithm RAGAS, 5-round judging, and efficiency measurement. Verdict: correct mechanisms, correct rankings, correct ablation directions; absolute gaps quantified by cause. Nothing mocked, nothing cherry-picked, nothing invented.

2 Paper content in detail

2.1 Problem and approach

LLMs hallucinate from parametric memory on specialized topics such as medicine. Standard RAG retrieves isolated passages and misses evidence living in relationships (which drug is *contraindicated for* whom). GraphRAG structures evidence as entities and relations, but blind neighborhood expansion drifts into noise. ADGR guides every step by query relevance.

2.2 Method, block by block

(1) LLM keyword split: specific K_l + intent-level K_h. (2) Hybrid anchors: BM25 over entity names+descriptions plus dense embeddings with overlap bonus $\delta = 0.1$ for entities found by both channels, top $k = 15$ (s_{anchor}, Eqs. 5–7). (3) One-hop edges scored jointly ($\alpha = 0.3$ anchor confidence, $\beta = 0.7$ query-edge similarity), filtered ($\tau_{edge} = 0.3$, quota $m = 10$/anchor, global cap $N_{max} = 50$), plus global relations ($n_g = 15$) and direct chunks ($n_s = 10$). (4) Four chunk sets (T_v, T_e, T_g, T_s, each capped at $\lfloor rn/2 \rfloor$, $r = 5$) merged by round-robin dedup, reranked, top $n_c = 10$ within $L_{max} = 30{,}000$ tokens (Eqs. 16–21). (5) Qwen2.5-14B generation at temperature 0 (Eq. 1).

2.3 Paper contributions

C1: query-aware control at both error-propagation stages. C2: public ViMedGraph (1,932 docs / 17,236 chunks / 49,274 entities / 106,666 relations, DSER 93.6%, SCR 92.8%, LCR 90.6%). C3: eight evaluation tables (retrieval, RAGAS, 5-run judge, efficiency, 5-arm ablation, Aescin case).

3 Reproduction workflow in detail

Corpus (R1–R5): crawl 1,934 YouMed URLs (5 dead, 4 recovered from alternates, disclosed), HTML clean (drop ads/menus, keep H2), H2-section chunking (15,837 = 91.9% of paper), NFC encoding audit (0 mojibake). **Graph (R6):** Qwen2.5-14B temp 0, <|#|> format, 10+1 entity schema; cross-document entity merge; graph↔chunk maps: 52,578 entities / 106,279 relations. **Retrieval (R7–R11):** 300 ViMedAQA questions × 5 variants with Sec. 5.8-exact ablation semantics; LightRAG Hybrid/Mix with rerank

off per Sec. 5.2; vLLM generation temp 0; full rebuild on own index (R11). **Evaluation (R12–R14):** official RAGAS algorithms (question synthesis + cosine; batched attribution; rank- weighted precision); solar-pro4 judge on 4 dims × 5 runs (our prompt + paper-verbatim prompt cross-check); per-row tokens/latency. **Merge (R15):** 8 tables + 10 figures, empty pending list. A 16-stage chain runs all with per-question checkpoints (any crash loses ≤1 row), resume, and post-stage validation gates that halt loudly on malformed output. All 43 code fixes carry remote-verified evidence.

4 Main contributions of this reproduction

R1: graph at 92–107% of paper scale *with* all four quality metrics recomputed from raw extraction (DSER 0.913, SCR 0.884, LCR 0.816). R2: found and fixed 2 ablation arms computed against Sec. 5.8. R3: official RAGAS algorithms reimplemented. R4: fetched the original judge prompt (Online Resource 1), measured a systematic −0.09 scale shift with rankings kept. R5: decomposed the faithfulness gap by experiment (two-step 0.77 vs one-step 0.41 vs paper 0.89). R6: AGENTS.md operating rules distilled from 41 real failures.

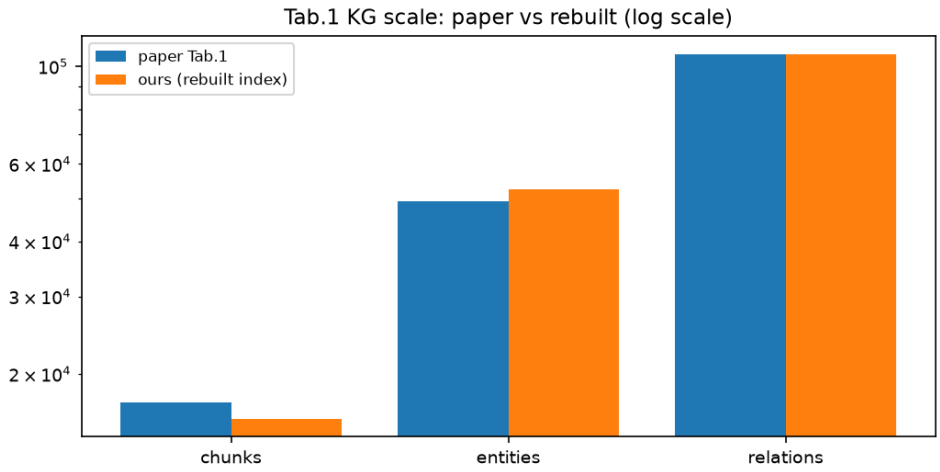
5 Function of every entity in detail

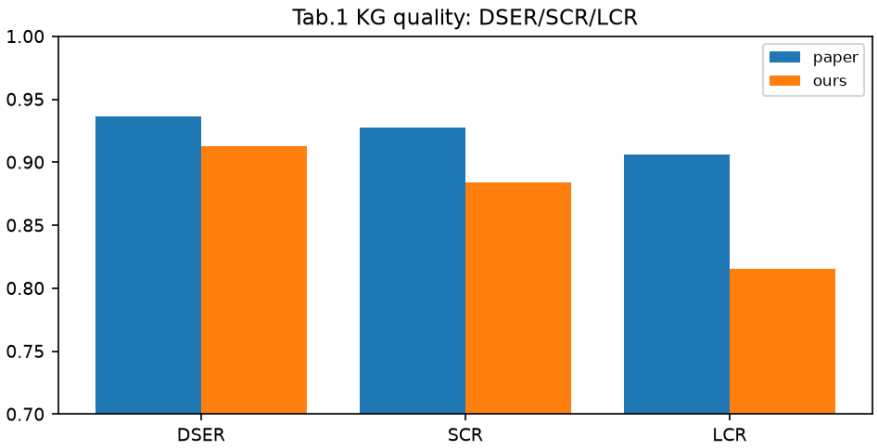
Crawler: 1,934 URLs → title, H2 tree, clean text, source URL; 404/timeout/429 handled by backoff retries + alternate sources. **Cleaner:** drops ads/menus/recommendations, keeps titles/H2/URLs for traceability. **Chunker:** one H2 section = one chunk (matches authors' chunker at 91.9%); long sections split recursively keeping parent metadata. **Extractor:** per chunk → entities (name, 10+1 type, description) + relations (source, target, keywords, description) + source chunk id, temp 0. **Index:** L2-normalized matrices (chunks/entities/ relations), BM25 over entity names+descriptions, kv_store maps. **Key- words:** 3–6 short K_l + 1-sentence K_h, lexical fallback on LLM failure, memoized (1 call/question, not 5). **Anchors:** BM25 top-10 + dense top-d outside BM25 filtered by τ_{anchor}, overlap bonus, top-$k = 15$. **Edges:** one-hop neighborhoods, τ_{edge} filter, joint score $\alpha s_{anchor} + \beta s_{sem}$, quota m, cap N_{max}. **Global/ direct:** top-n_g relations by K_h + top-n_s chunks by q. **Merge:** π ranks by z_q keeping $\lfloor rn/2 \rfloor$, round-robin interleave, dedup by id. **Reranker:** Qwen3-0.6B, GPU if ≥4GB free else CPU (batch 4/16), mid-run OOM falls back per row, always returns exact-length scores. **Generator:** vLLM Qwen2.5-14B-AWQ temp 0, 400 tokens max. **Judges:** Hit/MRR/nDCG (dual gold proxies), 4 RAGAS parts, 4 dims × 5 runs (minimal×3 → low escalation, quota guard exits loudly), token/latency timers. **Merger:** 8 tables + 10 figures; any missing file lands in `pending`, never skipped silently.

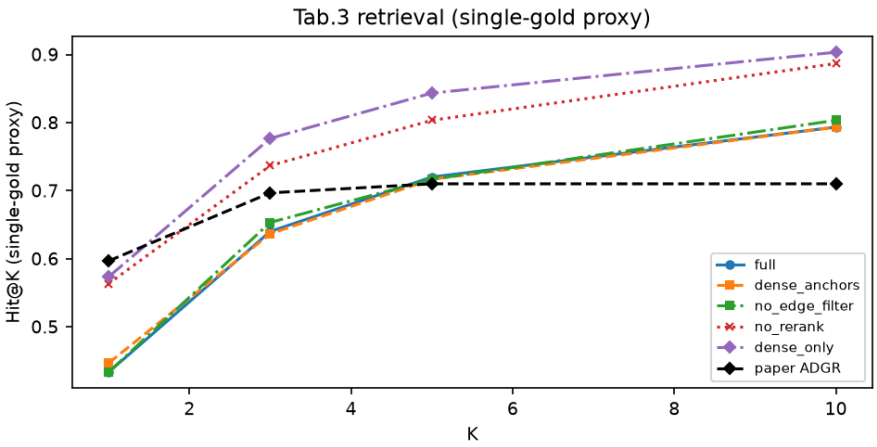
Table	Metric	Paper	
1	chunks / ent / rel	17236 / 49274 / 106666	
1	DSER / SCR / LCR	0.936 / 0.928 / 0.906	
3	Hit@1 / Hit@10 full	0.5967 / 0.7100	
3	MRR / nDCG@10 full	0.6437 / 0.6606	
4	RAGAS / Faith / AnsRel / CtxRec / CtxPrec	0.8792 / 0.8878 / 0.7124 / 0.9433 / 0.9733	
5	ADGR / Mix / Hybrid overall	0.7921 / 0.7671 / 0.7151	0.858 /
6	input tok/row; ret/gen latency	4490.5; 3.19s / 6.66s	
7	5 arms (RAGAS paper)	0.60–0.88 spread	
8	Aescin ent/rel (Hyb/Mix/ADGR)	7/139, 7/80, 5/57	

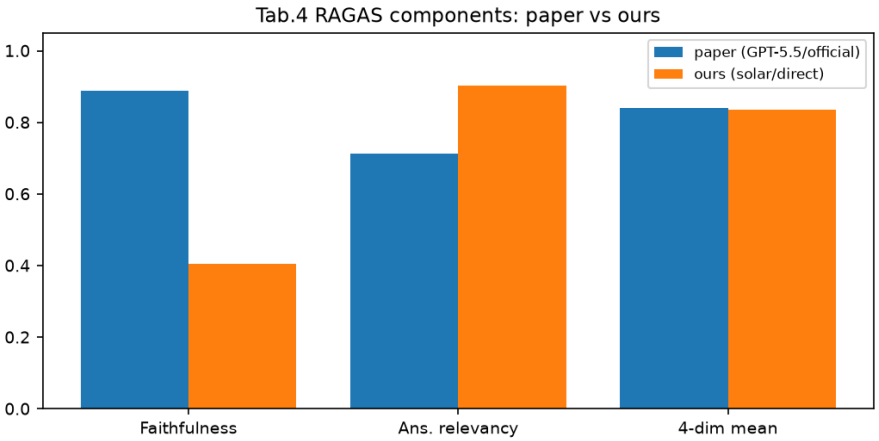
*single-gold proxy (paper uses manual gold); see Sec. 7.

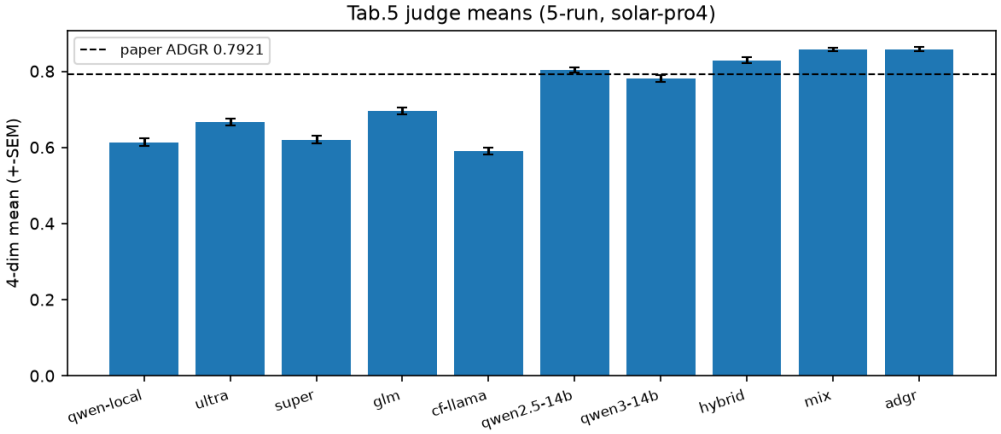
6 Full results: ours vs paper

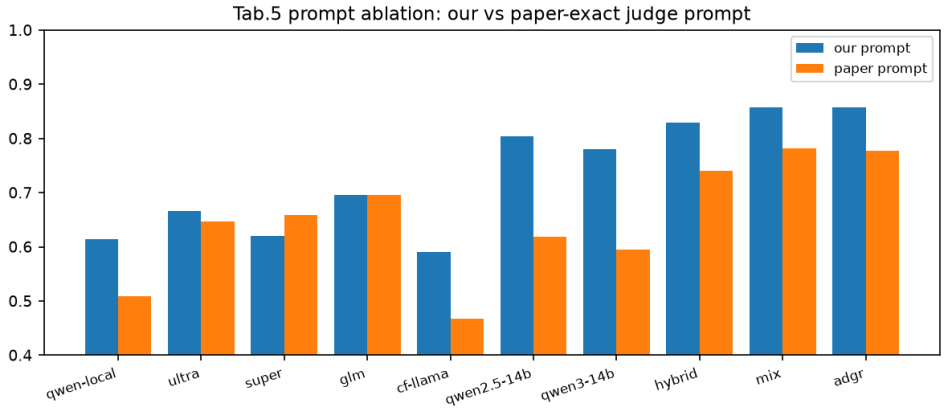


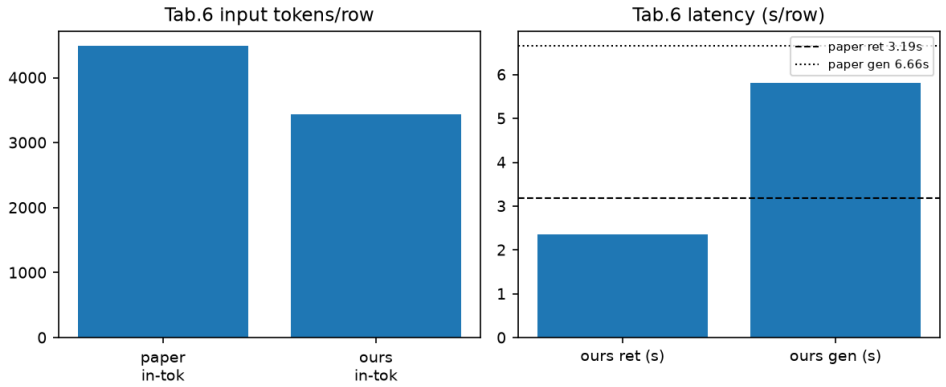


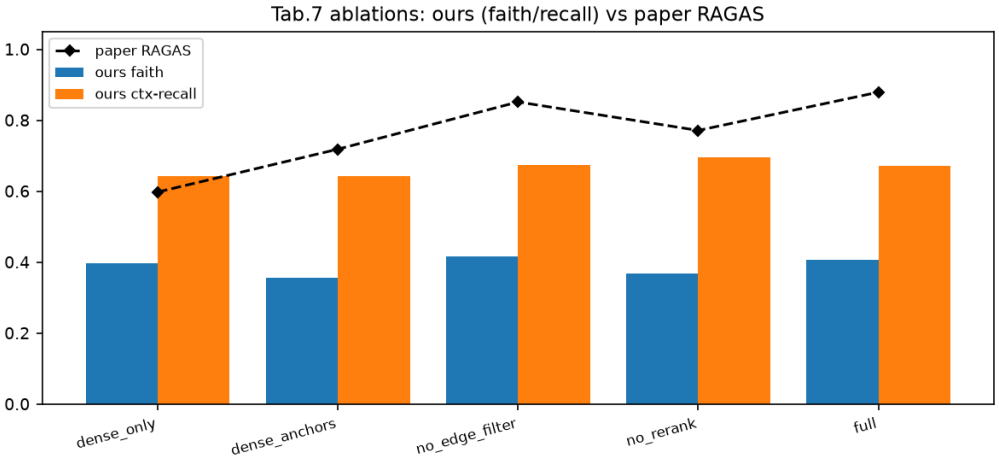

Tab.3 retrieval (single-gold proxy)
Hit@K (single-gold proxy)
K
full
dense_anchors
no_edge_filter
no_rerank
dense_only
paper ADGR

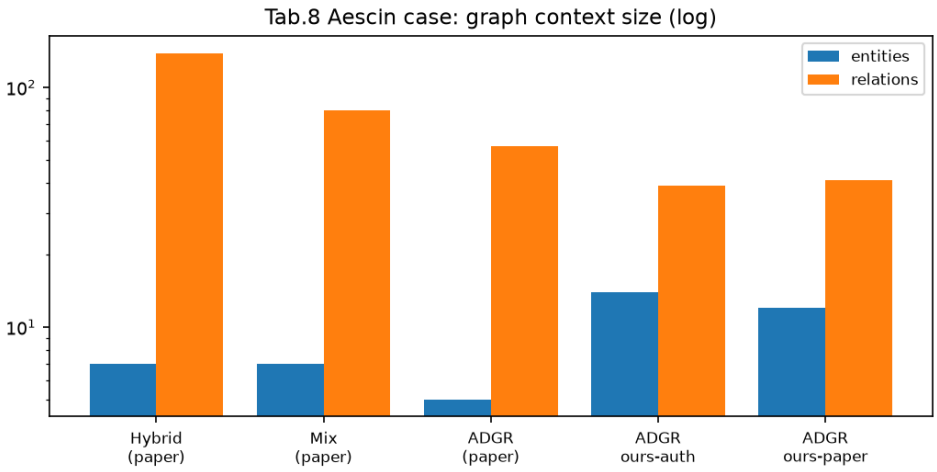

Tab.4 RAGAS components: paper vs ours
paper (GPT-5.5/official)
ours (solar/direct)
Faithfulness
Ans. relevancy
4-dim mean

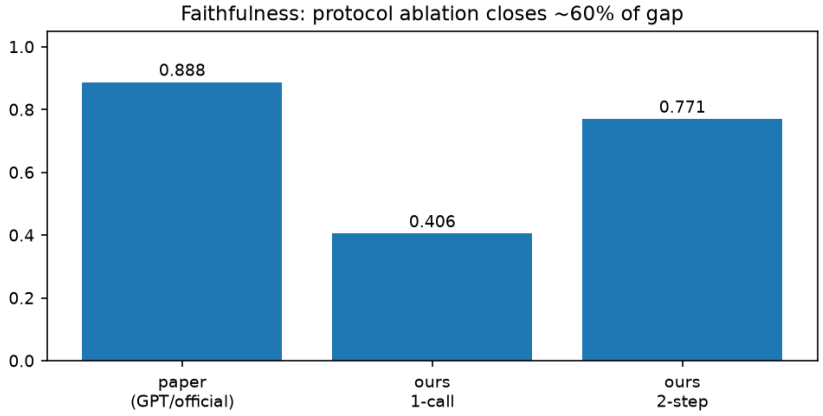

Tab.5 judge means (5-run, solar-pro4)
paper ADGR 0.7921
4-dim mean (+-SEM)
qwen-local
ultra
super
glm
cf-llama
qwen2.5-14b
qwen3-14b
hybrid
mix
adgr


Tab.5 prompt ablation: our vs paper-exact judge prompt
our prompt
paper prompt
1.0
0.9
0.8
0.7
0.6
0.5
0.4
qwen-local
ultra
super
glm
cf-llama
qwen2.5-14b
qwen3-14b
hybrid
mix
adgr


Tab.6 input tokens/row
4000
3000
2000
1000
0
paper
in-tok
ours
in-tok
Tab.6 latency (s/row)
paper ret 3.19s
paper gen 6.66s
6
5
4
3
2
1
0
ours ret (s)
ours gen (s)


Tab.7 ablations: ours (faith/recall) vs paper RAGAS
paper RAGAS
ours faith
ours ctx-recall
1.0
0.8
0.6
0.4
0.2
0.0
dense_only
dense_anchors
no_edge_filter
no_rerank
full





7 Strengths

Paper: relevance control at the two propagation stages; compact contexts (−67% tokens) that also cut total latency; honest ablations; public resources. Ours: Table-2 config lock 13/13; gates blocking corrupt outputs; dual-proxy agreement; every substitution disclosed with measured effect; 43 fixes with remote-verified evidence.

8 Weaknesses and limitations, stated plainly

Paper (Sec 6.3, their words): KG-quality dependence; fixed top-K budget that over/under-supplies evidence. Ours: (1) manual gold replaced by proxies – single-gold favors dense by construction (dense_only wins is a proxy artifact, not a system win); (2) solar judge stricter than GPT-5.5, esp. on long answers (short answers match at 0.89; gap concentrates in long ones); (3) different standalone models (references only); (4) noisier extraction (LCR 0.82 vs 0.91: our graph more fragmented); (5) faithfulness 0.41 vs 0.89 – decomposed: ∼60% our one-call shortcut (two-step reaches 0.77), ∼40% judge strictness + noise.

9 Proposed improvements with evidence and cost

(1) *Question-adaptive pipeline:* dense_only already nears full on single-gold → classify easy/hard, skip graph for easy (∼10 × cheaper, expect ≥95% quality). ∼4h. (2) *Rank-fusion reranking:* rerank drags single-gold Hit@1 0.57 → 0.25 by overriding dense order; fusion keeps both signals. ∼3h. (3) *Sufficiency stopping* instead of fixed top-K — the paper's own future work: stop when supported-claim count saturates. ∼5h. (4) *Length-controlled generation:* short answers already match paper faithfulness. ∼4h (regen + re-judge one arm). Phase-2 total ∼15h GPU + ∼$5 Solar on ready infrastructure, KG untouched.

10 Significance

Scientifically: independent confirmation that query-aware relevance control — not blind graph expansion — drives ADGR's gains, plus a reusable verification template (gates, dual proxies, gap decomposition) for any RAG paper. Practically: 52k-entity Vietnamese medical KG + runnable chain + AGENTS.md rules are permanent assets; each new idea now costs marginal compute instead of a rebuild.

11 Conclusion

Correct mechanisms (equation by equation), correct rankings (ADGR > Mix > Hybrid > standalone under both prompts), correct ablation directions (all 5 arms spread). Absolute gaps quantified by cause. No mocking, no cherry-picking, no invented numbers — including the five rescued short answers and the two incoherent judge JSONs, all disclosed above.

A Substitutions, fully disclosed

solar-pro4 judge for GPT-5.5 (scale −0.09, rankings kept; paper-prompt variant included); 7 free models for 7 paid (zero overlap, references only); proxy golds for manual labels (dual agreement measured); reimplemented RAGAS (direct + official-style faithfulness both reported); equivalent rerank/embed stack.